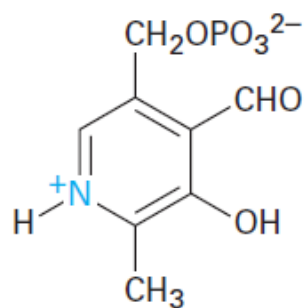
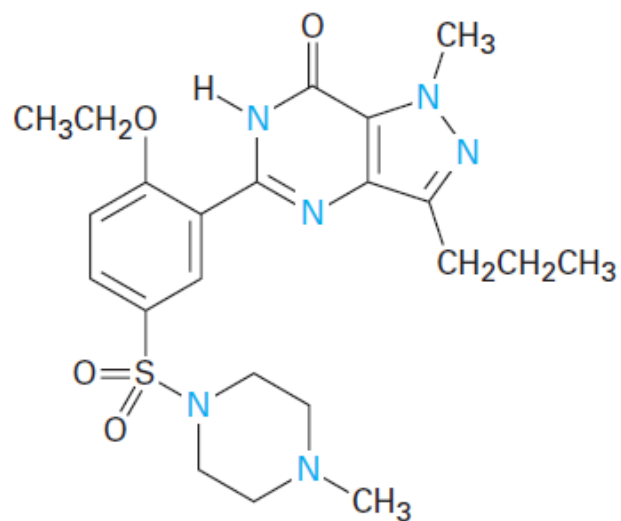


Heterocycles

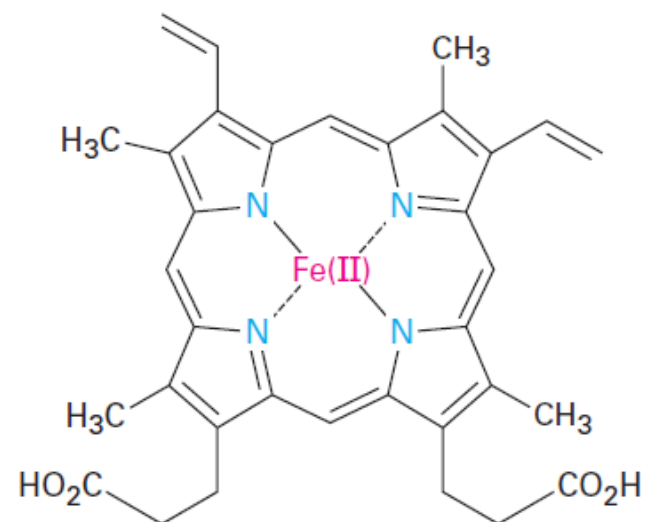
Yonatan A.



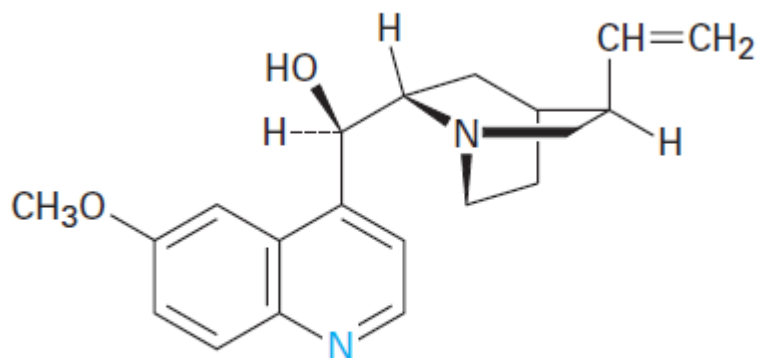
Pyridoxal phosphate
(a coenzyme)



Sildenafil
(Viagra)



Heme



Quinine
(antimalarial)

Learning objectives

- Define and differentiate the various classes and sub-classes of heterocycles
- Identify the chemical structures of heterocycles as they occur in drugs, natural products and other biologically important molecules.
- Discuss the inductive and resonance effects on the aromaticity, acid –base properties and reactivity of heterocycles.
- Describe the reactions of heterocycles.

Outline

- **Heterocycles**
- **Non-aromatic heterocycles**
- **Aromatic heterocycles**
 - **Aromaticity and heteroaromaticity**
 - **Six-membered aromatic heterocycles**
 - **Five-membered aromatic heterocycles**
 - **Six-membered rings with two heteroatoms**
 - **Five-membered rings with two heteroatoms**
 - **Heterocycles fused to a benzene ring**
 - **Fused heterocycles**

Detailed outline

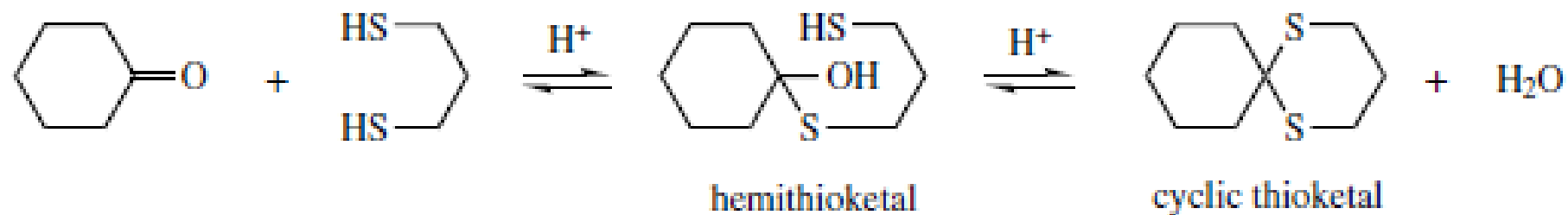
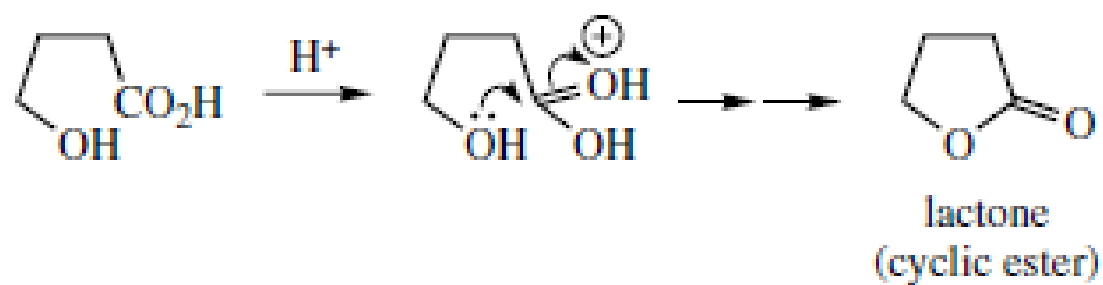
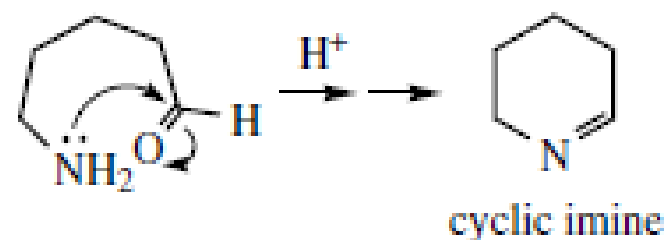
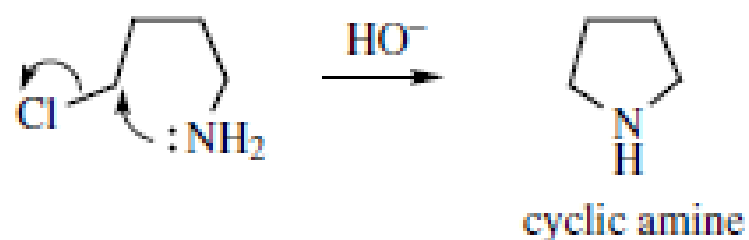
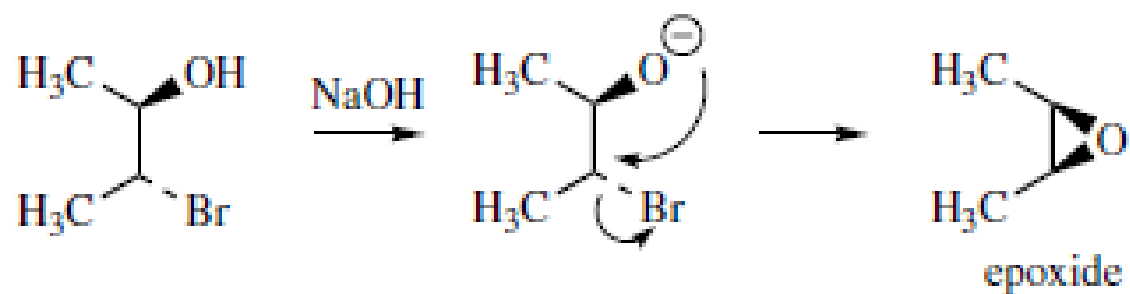
- **Heterocycles**
- **Non-aromatic heterocycles**
 - **Nomenclature and Numbering**
- **Aromatic heterocycles**
 - **Aromaticity and heteroaromaticity**
 - **Pyridine and pyrrole**
 - **Six-membered aromatic heterocycles**
 - **Pyridine, pyrylium cation and pyrones**
 - **Five-membered aromatic heterocycles**
 - **Pyrrole (azacyclopentadiene), furan and thiophene**
 - **Six-membered rings with two heteroatoms**
 - **Diazines**
 - **Five-membered rings with two heteroatoms**
 - **Imidazole, oxazole and thiazole**
 - **Heterocycles fused to a benzene ring**
 - **Quinoline, isoquinoline and indole**
 - **Fused heterocycles**
 - **Purines**

Heterocycles

- Cyclic compounds in which one or more of the ring atoms is not carbon are termed **heterocycles**; the noncarbon atoms are referred to as **heteroatoms**.
- The heteroatoms are nitrogen, oxygen, or sulfur.
- two classes, i.e. non-aromatic and aromatic.

Non-aromatic heterocycles

- many examples of nonaromatic heterocycles
 - cyclic ethers , including epoxides
 - cyclic amines, lactones, lactams and
 - cyclic acetals and ketals .
- the standard approach to generating heterocyclic systems requires a difunctional compound containing a leaving group or electrophilic centre, together with a nucleophilic species that provides the heteroatom.



Nomenclature

- the most important of these structures tend to have a trivial rather than systematic name, a consequence of long-standing common usage.
 - e.g. tetrahydrofuran and tetrahydropyran, are derived from the name of the corresponding aromatic heterocycle by the concept of reduction.

Numbering

- **Numbering** always begins at the heteroatom;
 - in the case of morpholine, numbering starts at oxygen, the heteroatom of higher atomic number.
- Remember that an accepted alternative in nomenclature is to indicate a heteroatom by the prefix aza-, oxa, or thia in the appropriate carbocycle.
 - piperidine as azacyclohexane,
 - and tetrahydrofuran as oxacyclopentane.

Non-aromatic heterocycles

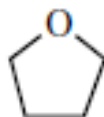
commonly used names shown in bold



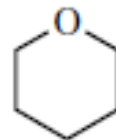
oxirane
(ethylene oxide)



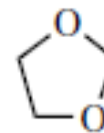
oxetane



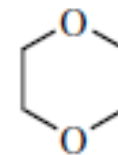
oxolane
(tetrahydrofuran)



oxinane
(tetrahydropyran)



1,3-dioxolane



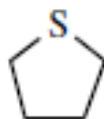
1,4-dioxane



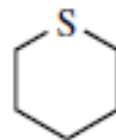
thiirane
(ethylene sulfide)



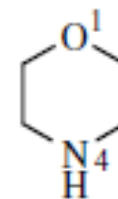
thietane



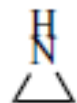
thiolane
(tetrahydrothiophene)



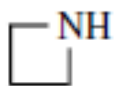
thiinane
(tetrahydrothiopyran)



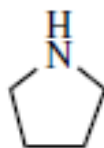
morpholine



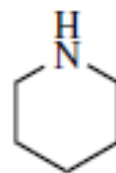
aziridine
(ethylene imine)



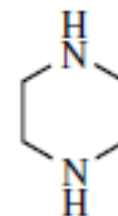
azetidine



azolane
(pyrrolidine)

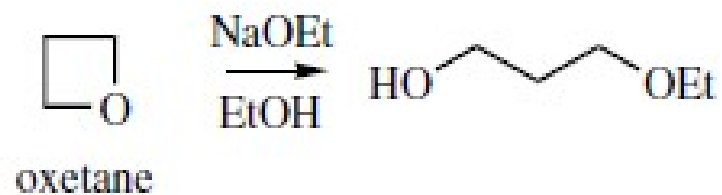
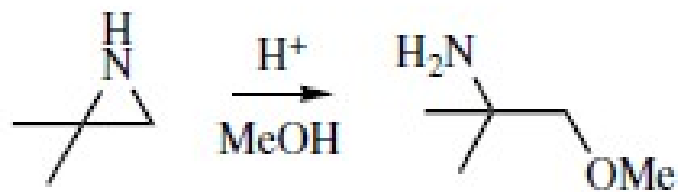
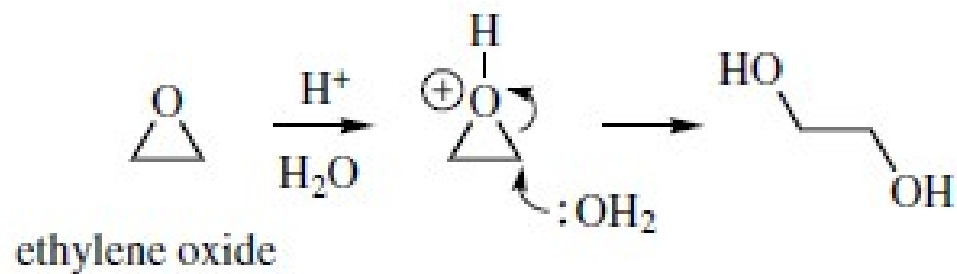


azinane
(piperidine)



piperazine

- we emphasize only the relative reactivity of the three-membered ring systems towards ring opening
 - Eg. epoxides, and similar reactivity is found with aziridines and thiiranes.
- Four-membered systems are also considerably strained and reactive towards nucleophiles, though not as readily as the three-membered compounds.



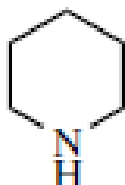
Use

- Some of these heterocycles provide us with valuable laboratory solvents, e.g. the ethers tetrahydrofuran and dioxane (1,4-dioxane).
- Others are useful as organic bases, e.g. piperidine, pyrrolidine, and morpholine.

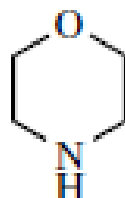
- The basicities of the nitrogen derivatives are comparable to those of similar acyclic amines, but physical properties, e.g. higher boiling point, make them more versatile than the simple amines.



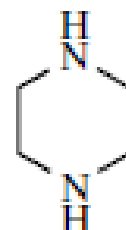
pyrrolidine



piperidine



morpholine



piperazine



diethylamine

pK_a
(conjugate acid)

11.3

11.1

8.5

9.7, 5.3

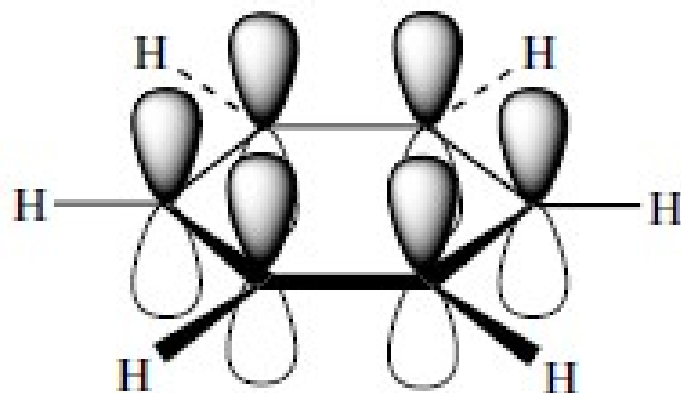
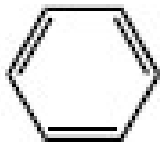
10.8

Aromatic heterocycles

Aromaticity and heteroaromaticity

- **Pyridine** is structurally related to benzene: one CH unit has been replaced by N.
- The ring atoms in benzene are sp^2 hybridized . The remaining singly occupied p orbitals are oriented at right angles to the plane of the ring, and overlap to form a delocalized π system, extending to form a closed loop above and below the ring.
- Compared with what we might expect for the hypothetical cyclohexatriene, this results in a considerable stabilization, with significantly modified structure and reactivity in benzene. We termed this **aromaticity**

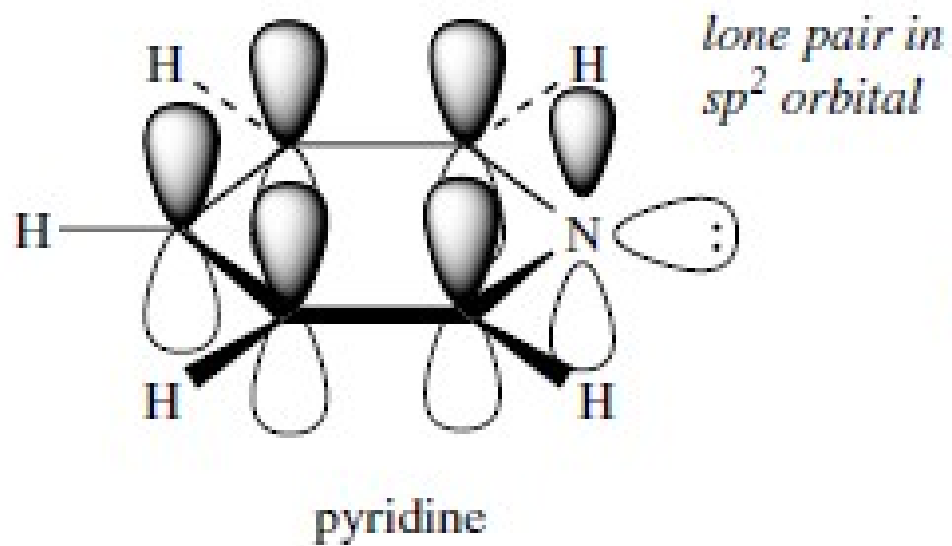
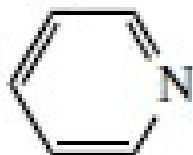
- Benzene conforms to **Hückel's rule**, which predicts that planar cyclic polyenes containing $4n + 2 \pi$ electrons show enhanced stability associated with aromaticity

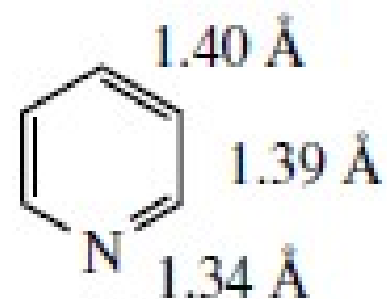
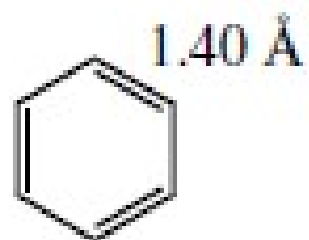


benzene

Pyridine

- Pyridine is also aromatic: nitrogen contributes one electron in a p orbital to the π electron system, and its lone pair is located in an sp^2 orbital that is in the plane of the ring and perpendicular to the π electron system.
- It also conforms to Hückel's rule, in that we still have an aromatic sextet of π electrons.



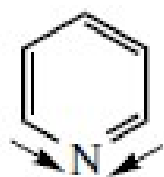


bond lengths in benzene and pyridine

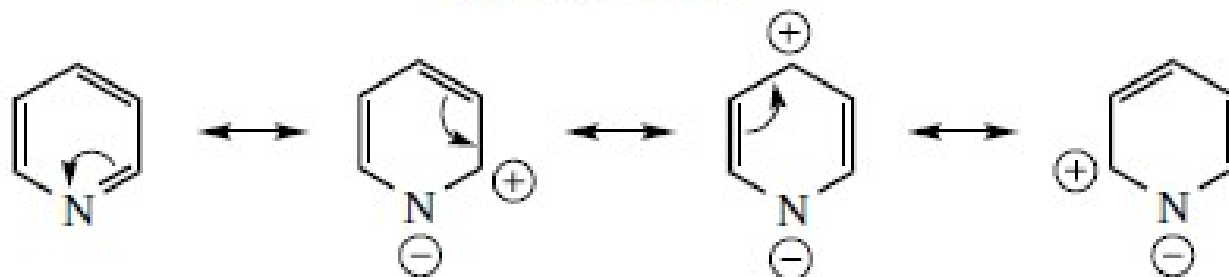
- One of the structural features of benzene that derives from aromaticity is the equal length of the C–C bonds ($1.40 \text{ \AA}$), which lies between that for normal single ($1.54 \text{ \AA}$) and double ($1.34 \text{ \AA}$) bonds.
- Pyridine does not have a perfect hexagon shape; the symmetry is distorted because the C–N bonds are slightly shorter ($1.34 \text{ \AA}$) than the C–C bonds ($1.39\text{--}1.40 \text{ \AA}$).

- Nitrogen is more electronegative than carbon, and this influences the electron distribution in the π -electron system in pyridine through inductive effects, such that nitrogen is electron rich.
- In addition, the nitrogen will also become electron rich through a resonance effect: several resonance forms may be drawn that have a negative charge on nitrogen.
- These effects thus reinforce each other.
- The heteroatom thus distorts the π electron cloud of the aromatic ring system, drawing electrons towards the nitrogen and away from the carbons.

inductive effect



resonance effect



*resonance effect
reinforces
inductive effect*



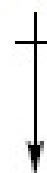
dipole



*dipole is
substantially
greater than
in piperidine*



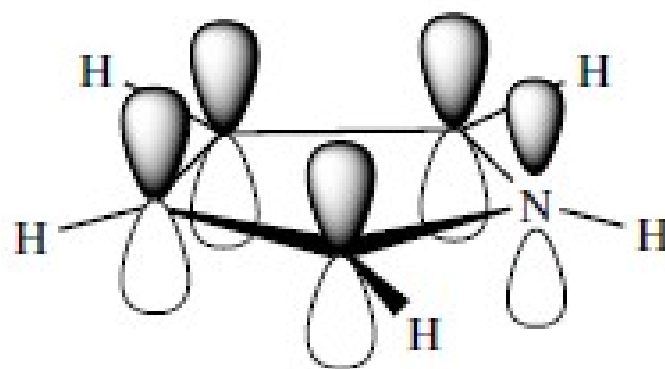
dipole



- An experimental probe for aromaticity is the chemical shift of the hydrogen signals in NMR spectroscopy.
- The substantially greater δ values for benzene protons (δ 7.27 ppm) compared with those in alkenes (δ 5–6 ppm) have been ascribed to the presence of a **ring current** that creates its own magnetic field opposing the applied magnetic field.
- This ring current is the result of circulating electrons in the π system of the aromatic ring.

Pyrrole

- Five-membered ring containing nitrogen.
- Pyrrole is also aromatic.
- How can we get six π electrons from just five atoms?
 - Each carbon contributes one electron as before, but nitrogen now contributes two electrons, its lone pair, to the π electron system.



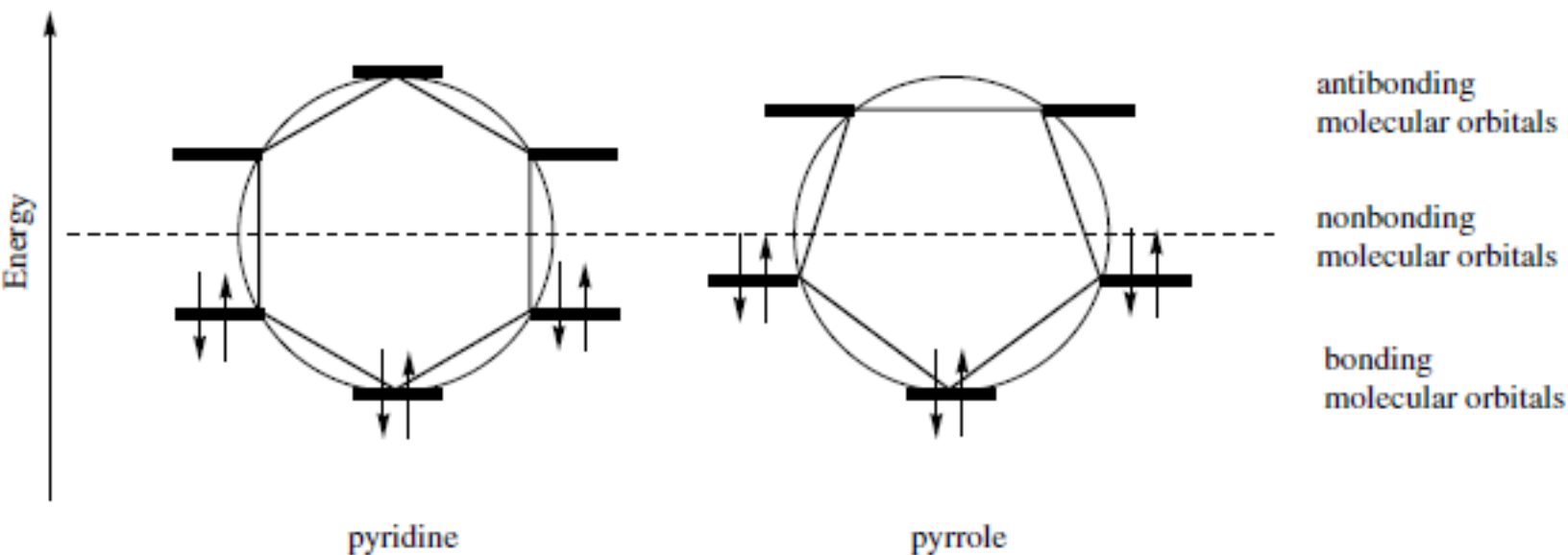
pyrrole

Energy diagrams of pyridine and pyrrole

- Relative energies of pyridine and pyrrole molecular orbitals from Frost circles

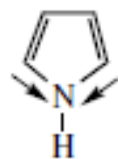
SIX-MEMBERED AROMATIC HETEROCYCLES

407

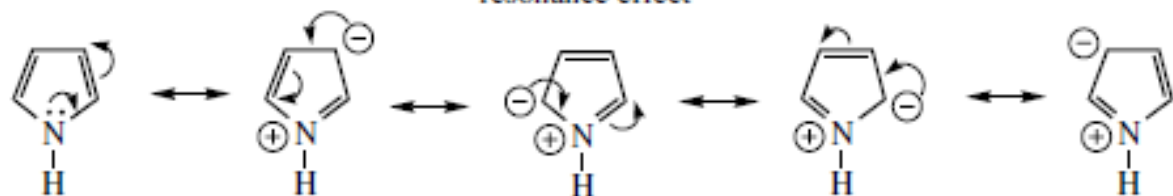


- However, the contribution of the nitrogen lone pair to the aromatic sextet in pyrrole makes the nitrogen atom relatively electron deficient.
- The nitrogen atom should create an inductive effect, as in pyridine, drawing electrons towards the heteroatom.
- However, a consideration of the resonance structures leads to several resonance forms with a positive charge on nitrogen.
- The resonance effect is opposite to the inductive effect, and of greater magnitude.
- Overall, the heteroatom distorts the π electron cloud of the aromatic ring system by pushing electrons away from the nitrogen and towards the carbons.

inductive effect



resonance effect



- The difference in electron distribution in pyridine and pyrrole manifests itself via the measured **dipole moments**.
- electron distribution influences the chemical reactivity of the two systems.

*larger resonance
effect opposes
inductive effect*



dipole



*contrast
inductive effect
in pyrrolidine*



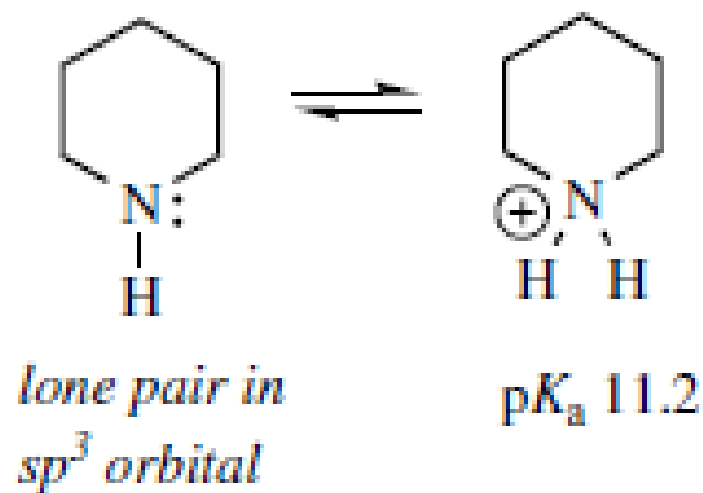
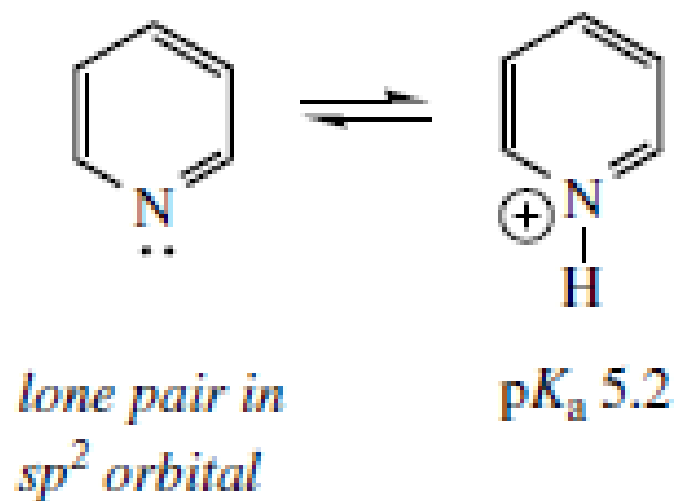
dipole



Six-membered aromatic heterocycles

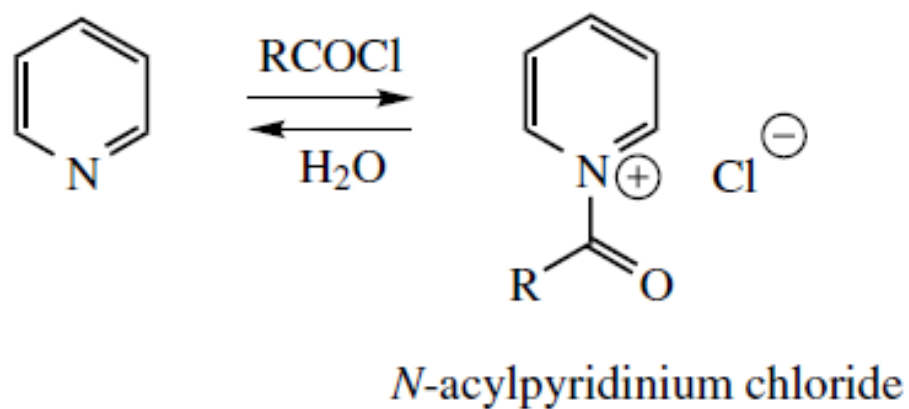
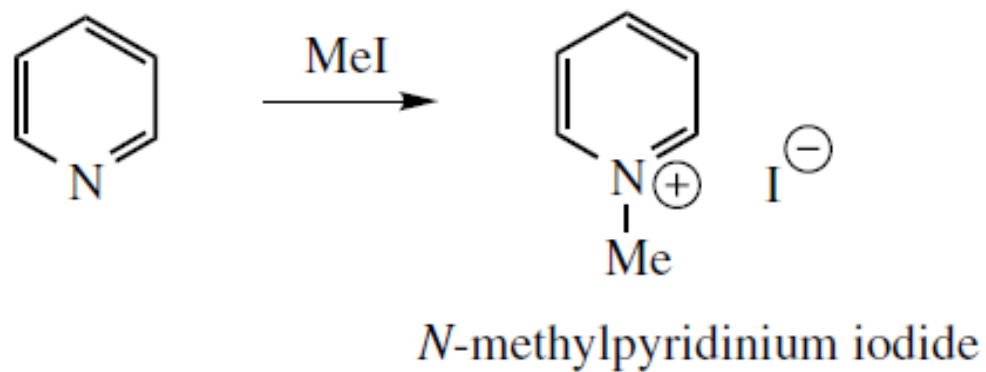
- **Pyridine** is a base (pK_a pyridinium cation 5.2), but it is a considerably weaker base than a typical non-aromatic heterocyclic amine such as piperidine (pK_a piperidinium cation 11.2).
- This is because the lone pair electrons in pyridine are held in an sp^2 orbital.
- The increased s character of this orbital, compared with the sp^3 orbital in piperidine, means that the lone pair electrons are held closer to the nitrogen, and are consequently less available for protonation.
- The lower basicity of pyridine compared with piperidine is thus a **hybridization effect** .

- Although pyridine is a weak base, it can form salts with acids and is widely used in chemical reactions as an acid scavenger and as a very good polar solvent.



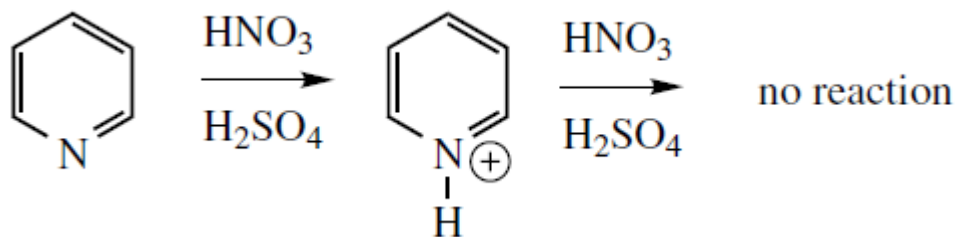
Reactivity of pyridine

- Just as pyridine is a weaker base than piperidine, it is also a poorer nucleophile.
- Nevertheless, it reacts with electrophiles to form stable **pyridinium salts**.
- In the examples shown, primary alkyl halides form *N*-alkylpyridinium salts, whereas acyl halides and anhydrides react to give *N*-acylpyridinium salts.



Electrophilic substitution

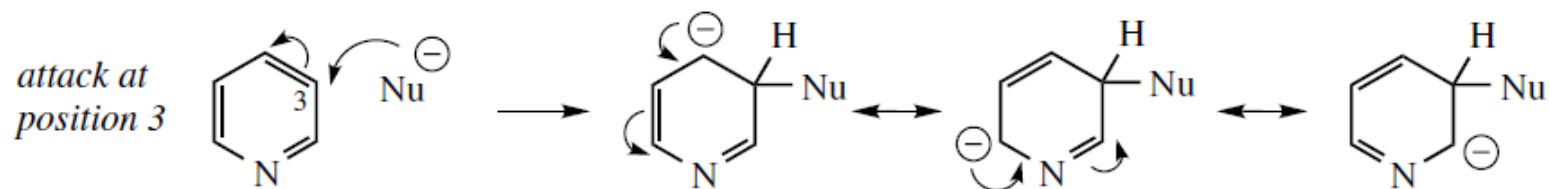
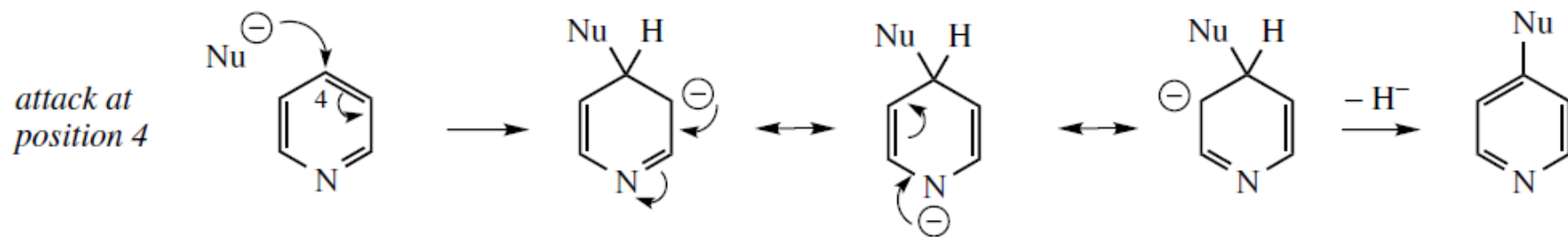
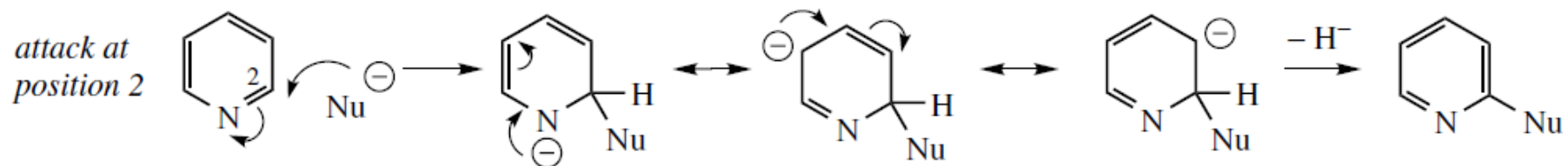
- The pyridine ring is rather unreactive towards electrophilic reagents.
- and these tend to be attacked by the nitrogen instead, making the ring even less reactive.



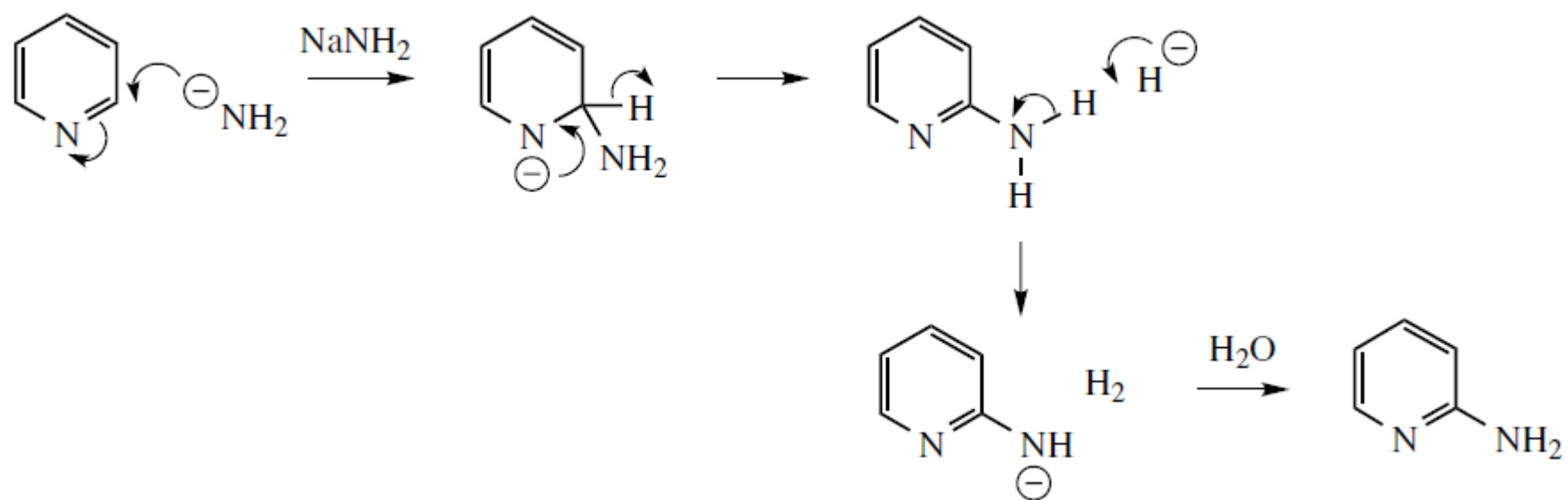
- A striking demonstration of the reduced activity towards electrophiles for the pyridine ring compared with the benzene ring will be seen later when we consider the fused heterocycles quinoline and isoquinoline .
- These contain a benzene ring fused to a pyridine ring; electrophilic substitution occurs exclusively in the benzene ring.

Nucleophilic aromatic substitution

- Pyridine, on the other hand, is more reactive than benzene towards **nucleophilic aromatic substitution**.
- This is effectively reaction towards the C=N ‘imine’ function, as described above.
- Attack is principally at positions 2 and 4, as predictable from resonance structures of reaction intermediates.
- Attack at the 3 position does not allow the nitrogen to help stabilize the negative charge.

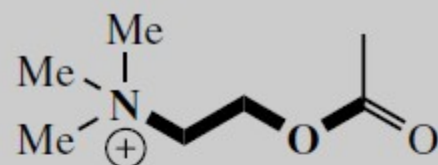
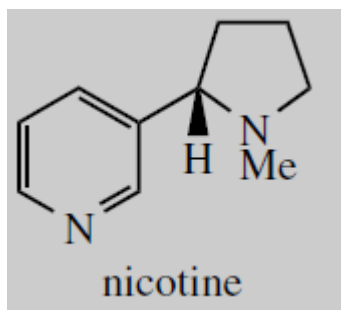


- However, for an unsubstituted pyridine, the leaving group to finish off this reaction is hydride, which is a strong base and thus a poor leaving group.
- It may be necessary to use an oxidizing agent to function as hydride acceptor to facilitate this type of hydride transfer.
- Nevertheless, there is a classic example of this process, known as the **Chichibabin reaction**, in which pyridine is converted into 2-aminopyridine through heating with sodium amide.

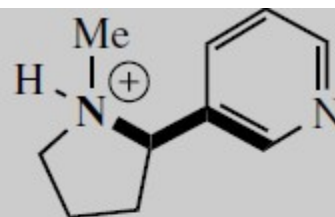


- The hydride released appears to abstract a proton from the product since the other product of the reaction is gaseous hydrogen.
- The aminopyridine anion is finally quenched with water.
- The product is mainly 2-aminopyridine, probably the result of the enhanced inductive effect on carbons immediately adjacent to the electronegative nitrogen.

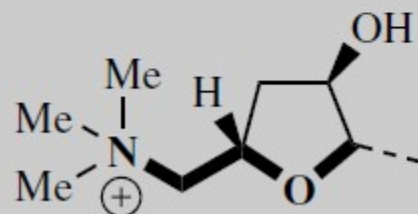
Nicotine, nicotinic acid, and nicotinamide



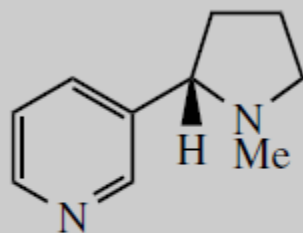
acetylcholine



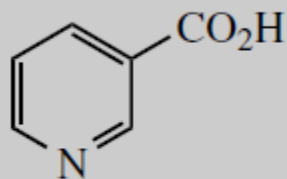
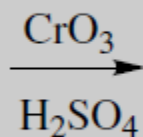
nicotine
(as conjugate acid)



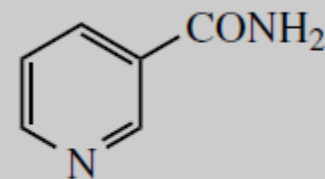
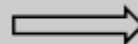
muscarine



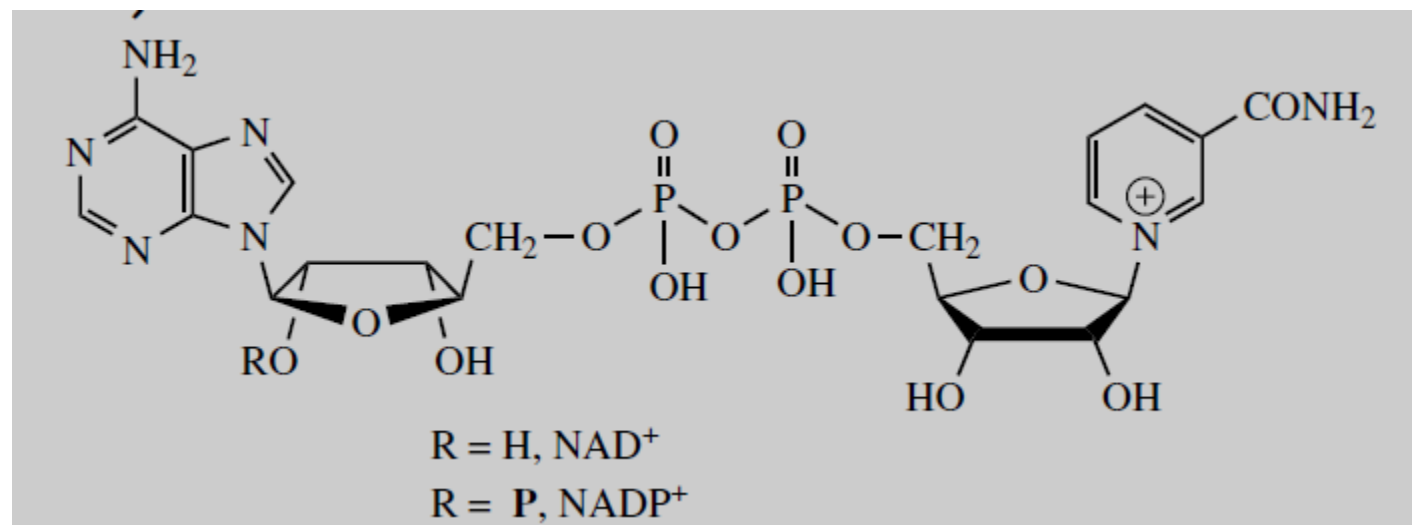
nicotine



nicotinic acid

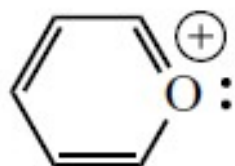


nicotinamide

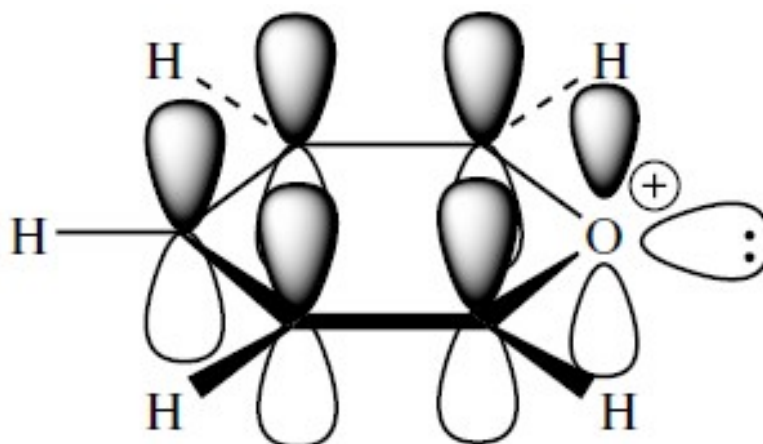


Pyrylium cation and pyrones

- The **pyrylium cation** is isoelectronic with pyridine: it has the same number of electrons and, therefore, we also have aromaticity.
- Oxygen is normally divalent and carries two lone pairs.
- If we insert oxygen into the benzene ring structure, then it follows that, by having one electron in a p orbital contributing to the aromatic sextet, there is a lone pair in an sp^2 orbital, and the remaining electron needs to be removed, hence the pyrylium cation.
- However, oxygen tolerates a positive charge less readily than nitrogen, and aromatic stabilization is less than with pyridine.

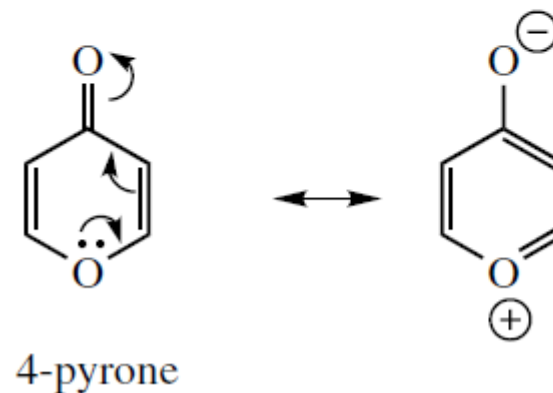
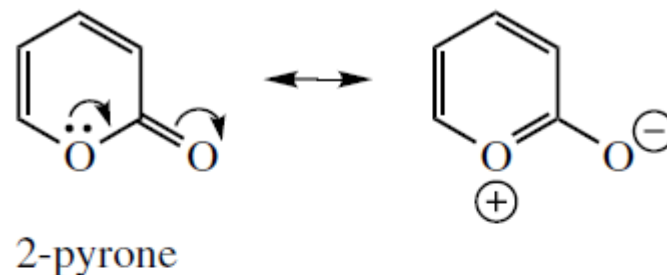


pyrylium cation

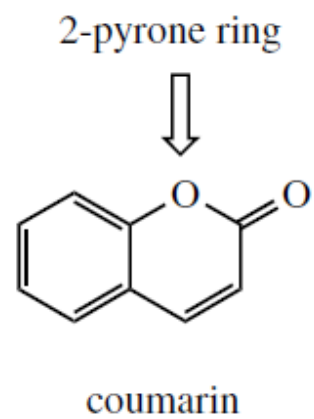
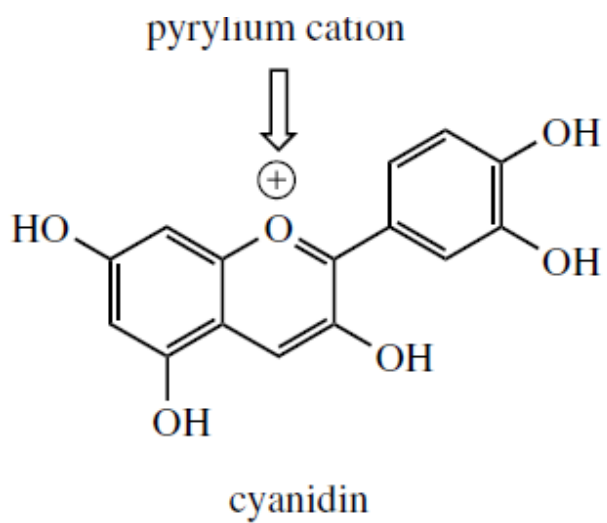
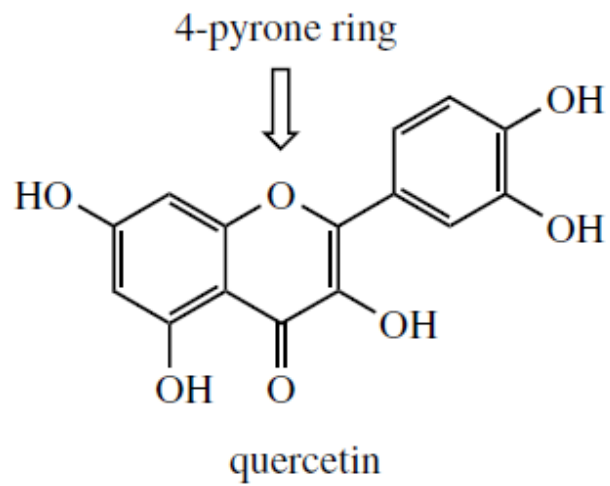


Pyrones

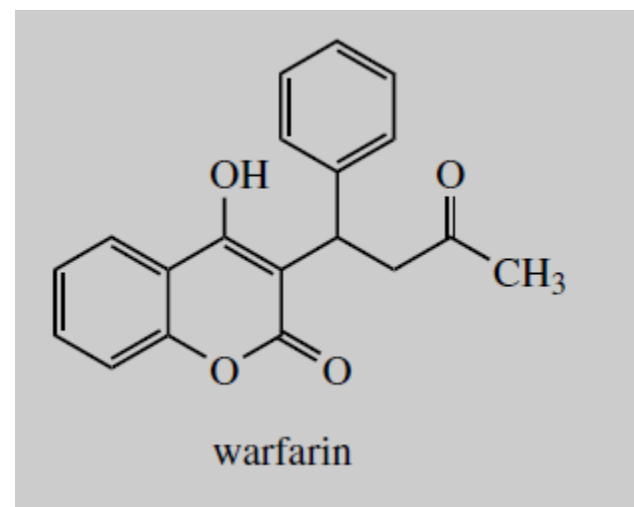
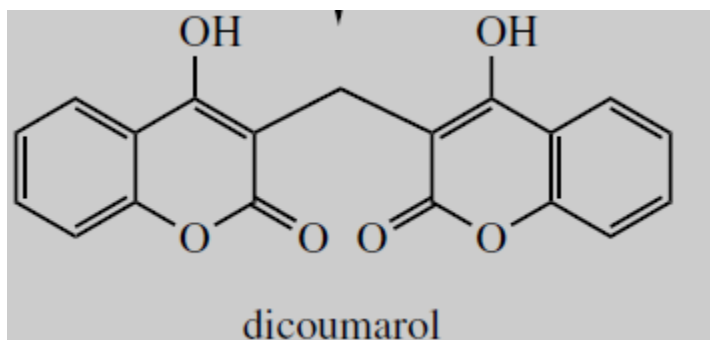
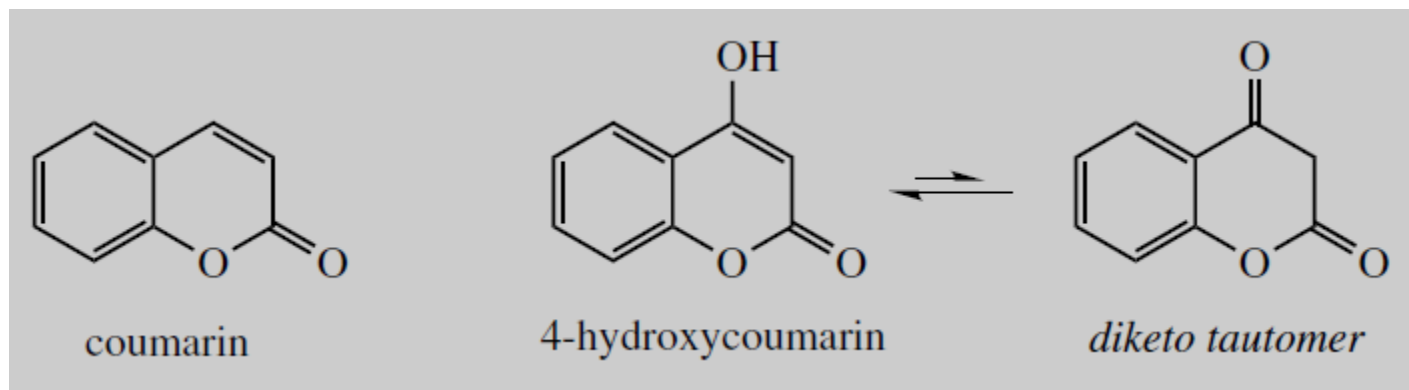
- Their chemical behaviour suggests they should be viewed more as conjugated lactones (2-pyrones) or vinylogous lactones (4-pyrones) rather than aromatic systems, since many reactions lead to ring opening.



- **Flavonoids** are natural phenolic systems containing pyrylium and pyrone rings, and provide the most prominent examples.
- **Coumarins** contain a 2-pyrone system. Note that all of these compounds are fused to a benzene ring and are strictly benzopyran or benzopyrylium systems.

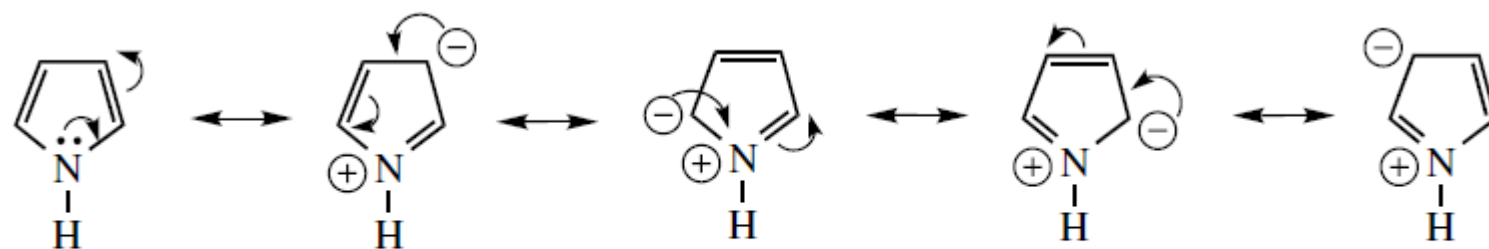


Dicoumarol and warfarin



Five-membered aromatic heterocycles

- **Pyrrole** (azacyclopentadiene)
 - is the ring system obtained if we replace the CH₂ group of cyclopentadiene with NH.
 - pyrrole has aromatic character because nitrogen contributes two electrons, its lone pair, to the π -electron system.
 - We have also noted from resonance forms that nitrogen carries a partial positive charge, and the carbons are electron rich.
 - This is stronger than the opposing inductive effect.

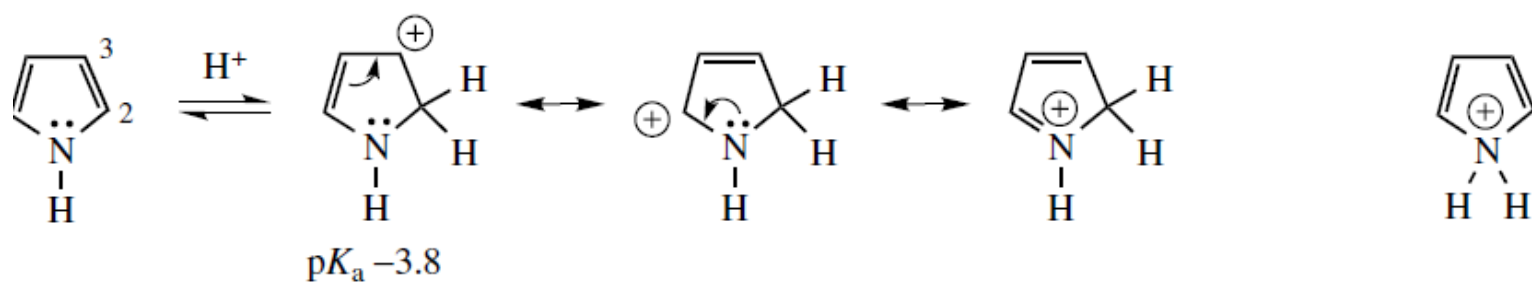


pyrrole

resonance forms predict nitrogen carries partial positive charge, and carbons are electron rich

Acid-base properties of pyrrole

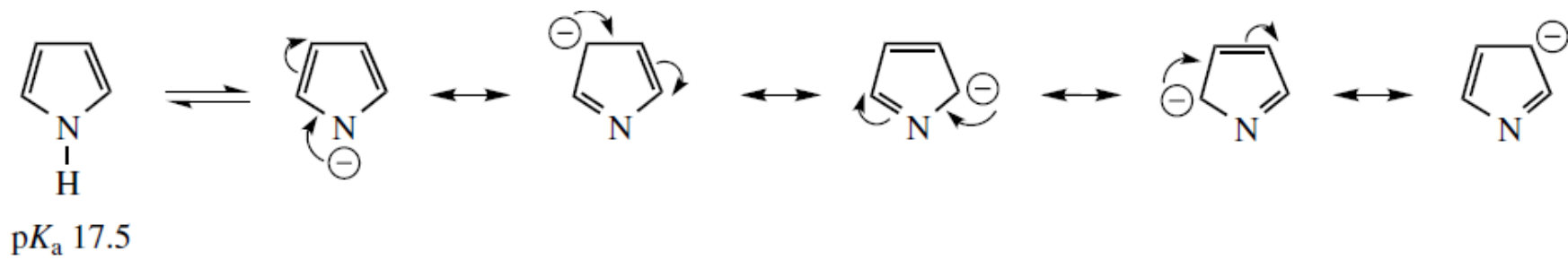
- Pyrrole is a particularly weak base, with pK_a of the conjugate acid -3.8 .
- First, we should realize that protonation of pyrrole will not occur on nitrogen: nitrogen has already used up its lone pair by contributing to the aromatic sextet, so protonation would necessarily destroy aromaticity.



*protonation on C-2; aromaticity destroyed,
but resonance stabilization of cation*

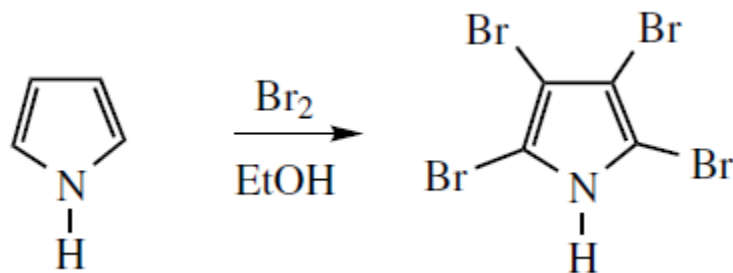
*protonation on N not favoured;
destroys aromaticity*

- It turns out that, as opposed to acting as a base, pyrrole is potentially an acid (pK_a 17.5);
- it is not a particularly strong acid, but stronger than we might expect for a secondary amine system (pK_a about 36).
- This is because the anion formed by losing the proton from nitrogen has a negative charge on the relatively electronegative nitrogen, but maintains its aromaticity.
- Unlike in pyrrole, the anion resonance structures do not involve charge separation.



Reactions of pyrrole

- it will react readily with electrophiles; or,
- put another way, pyrrole will behave as a nucleophile.
- Pyrrole is very reactive towards electrophiles. For example, treatment with bromine leads to substitution of all four positions.

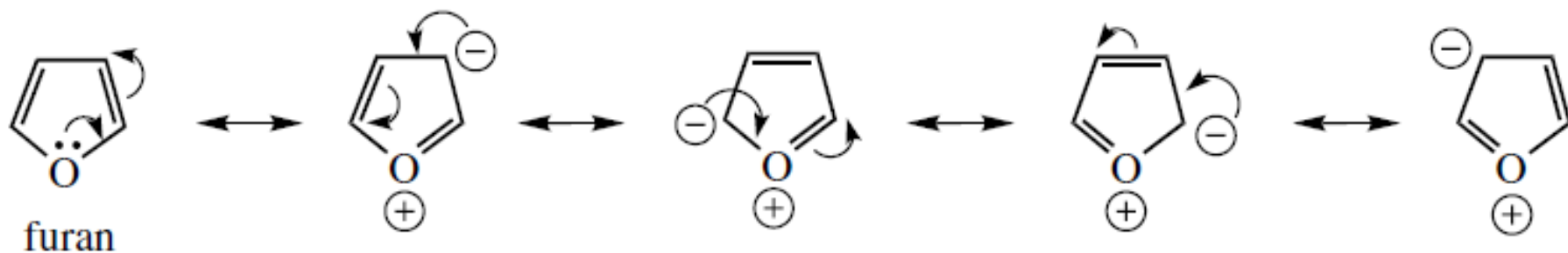


Furan and thiophene

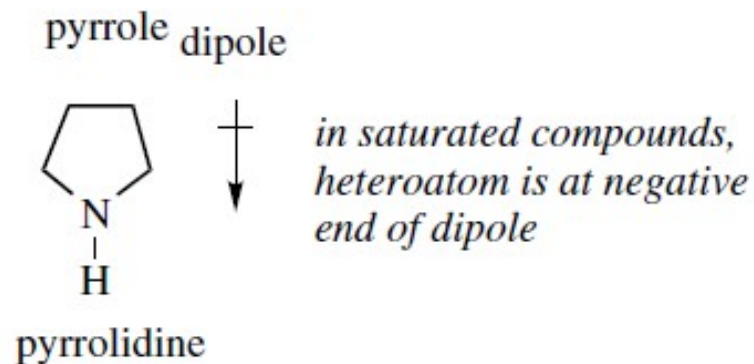
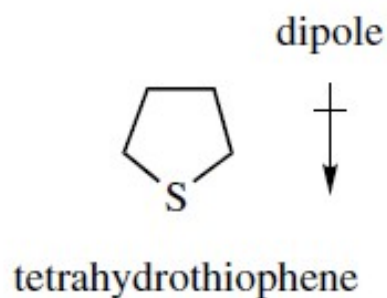
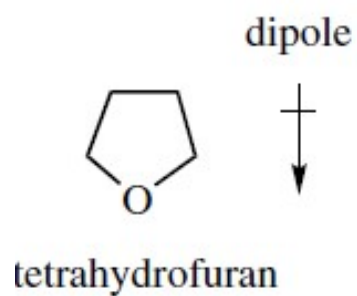
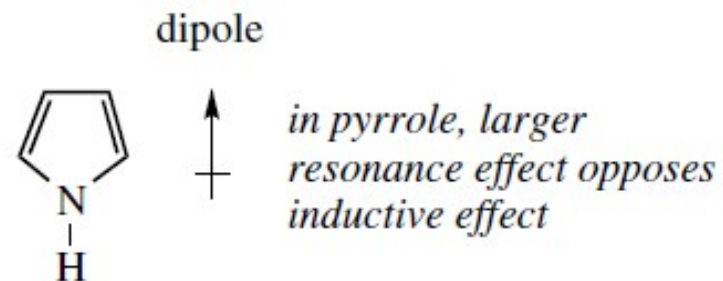
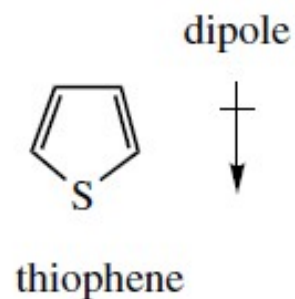
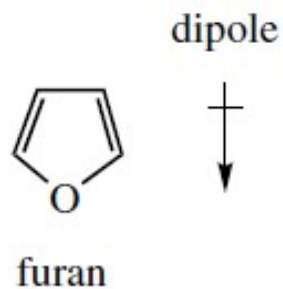
- **Furan** and **thiophene** are the oxygen and sulfur analogues respectively of pyrrole.
- Oxygen and sulfur contribute two electrons to the aromatic sextet, but still retain lone pair electrons.
- There is one significant difference, however, in that oxygen uses electrons from a $2p$ orbital, whereas the electrons that sulfur contributes originate from a $3p$ orbital.
- In the case of thiophene, this reduces orbital overlap with the carbon $2p$ orbitals.

- Both compounds are thus aromatic, and their chemical reactivity reflects what we have learnt about pyrrole.
- The most typical reaction is electrophilic substitution.
- However, we find that pyrrole is more reactive than furan towards electrophiles and thiophene is the least reactive; all are more reactive than benzene.
- This relates to the relative stability of positive charges located on nitrogen, oxygen and sulfur.
- We used similar electronegativity reasoning to explain the relative basic strengths of nitrogen, oxygen and sulfur derivatives .

- Furan is also the ‘least aromatic’ of the three, i.e. it has the least resonance stabilization, and undergoes many reactions in which the aromatic character is lost, e.g. addition reactions or ring opening.



- Note that the dipoles of furan and thiophene are opposite in direction to that in pyrrole.
- In furan and thiophene, there is a greater inductive effect opposing the resonance effect, whereas in pyrrole the resonance contribution was greater.
- In the non-aromatic analogues, the heteroatom is at the negative end of the dipole in all cases.

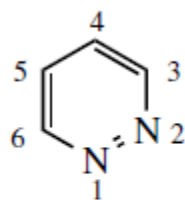


Six-membered rings with two heteroatoms

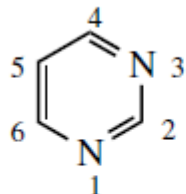
- **Diazines**

- diazabenzene, i.e. a benzene ring in which two of the CH functions have been replaced with nitrogen, is termed a **diazine**.
- Three isomeric variants are possible;
- these are called **pyridazine**, **pyrimidine**, and **pyrazine**.

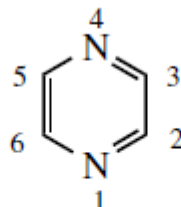
- These structures are all aromatic, the nitrogen atoms functioning in the same way as the pyridine nitrogen, each contributing one p electron to the aromatic sextet, with a lone pair in an sp^2 orbital.
- The diazines are much weaker bases than pyridine (pK_a 5.2).
- If we consider the inductive and resonance effects in pyridine, we have seen that these both draw electrons towards the nitrogen.



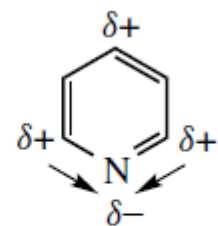
pyridazine
 pK_a 2.3



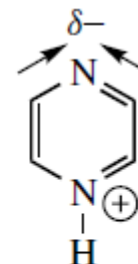
pyrimidine
 pK_a 1.3



pyrazine
 pK_a 0.7

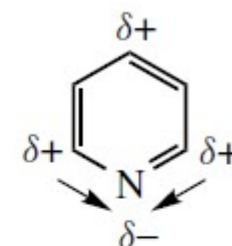
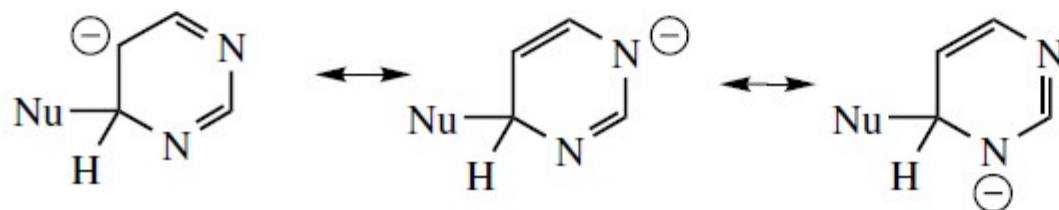
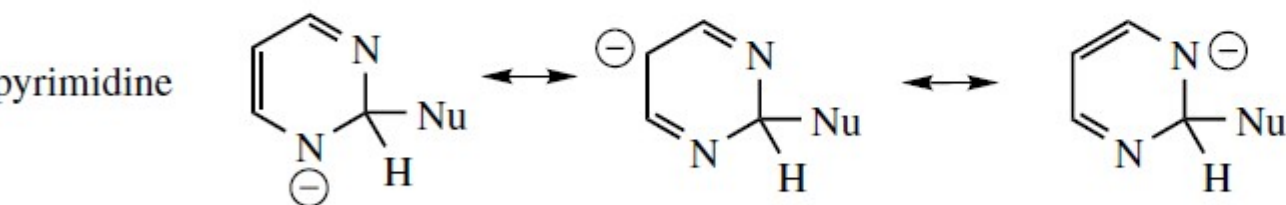
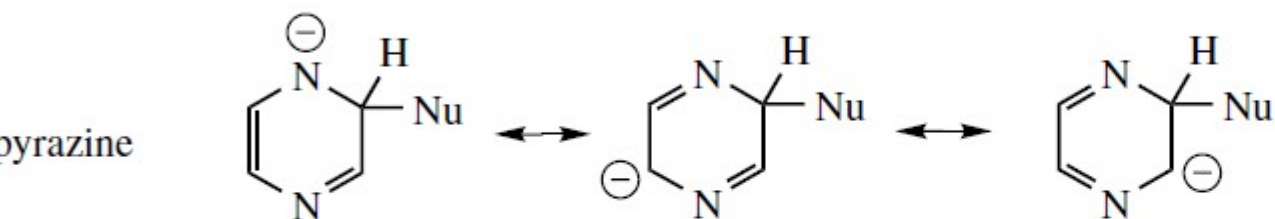
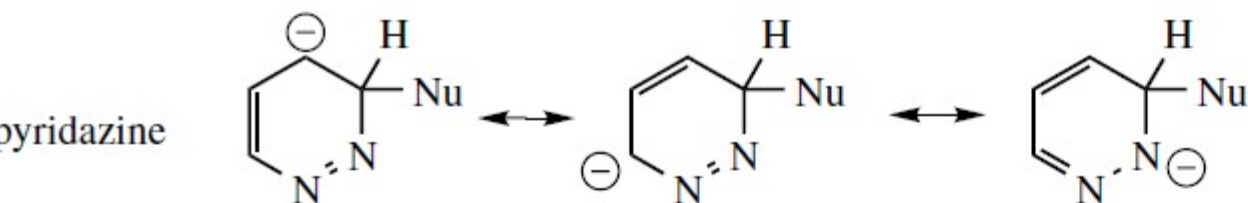


*inductive and
resonance effects
in pyridine*



*inductive effects in
pyrazinium cation*

- Diazines are more susceptible to nucleophilic attack than pyridine.



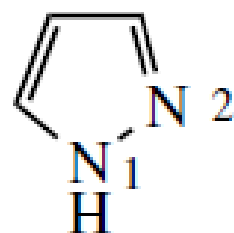
a second nitrogen will generate electron deficiencies at atoms where we are locating the negative charge

both nitrogens share the negative charge

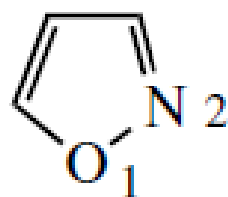
Five-membered rings with two heteroatoms

- In five-membered aromatic heterocycles with two heteroatoms, we can have
 - two nitrogens,
 - one nitrogen plus one oxygen,
 - or one nitrogen plus one sulfur.
- The heteroatoms can be positioned only 1,2 or 1,3.
- Numbering of the ring system starts from the heteroatom with the higher atomic number;

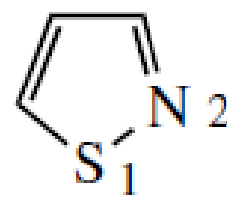
1,2-azoles



pyrazole

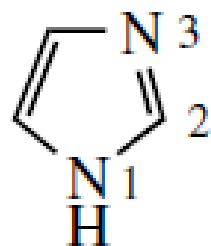


isoxazole

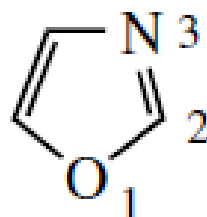


isothiazole

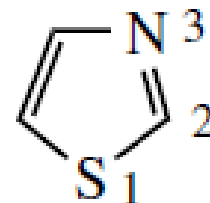
1,3-azoles



imidazole
1H-imidazole



oxazole

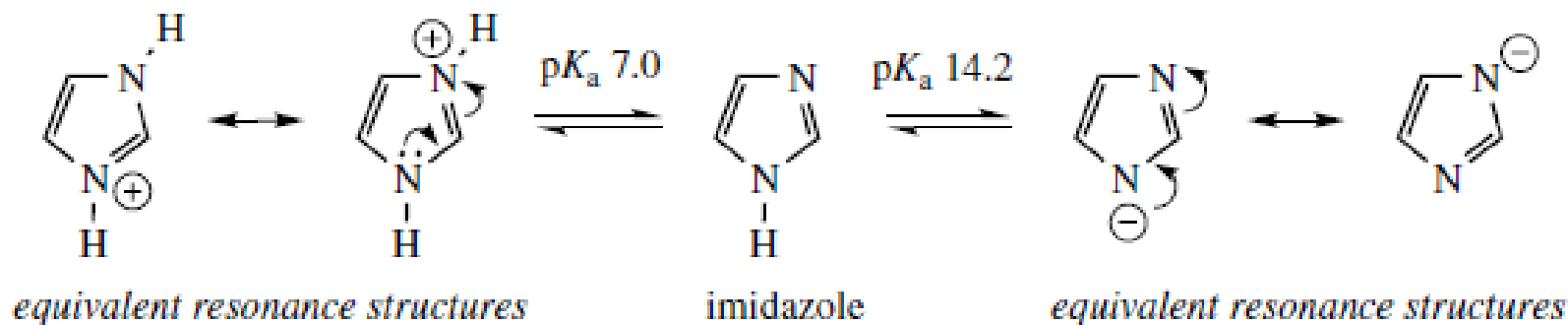


thiazole

Imidazole

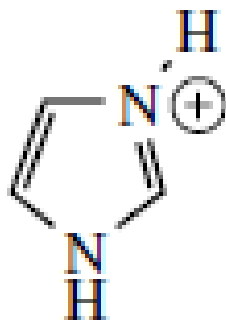
- Imidazole (pK_a of its conjugate acid 7.0) is a stronger base than either pyridine (pK_a 5.2) or pyrrole (pK_a – 3.8).
- The enhanced basicity of imidazole appears to stem from the
 - symmetry of the conjugate acid,
 - and the resonance stability conferred by this.
- The 1,3-relationship allows the two nitrogen atoms to share the charge equally.
- Note that pK_a 7.0 means imidazole is 50% protonated in water.

- Imidazole (pK_a 14.2) is also more acidic than pyrrole (pK_a 17.5); this, again, is a feature conferred by symmetry and the enhanced resonance stabilization in the conjugate base.

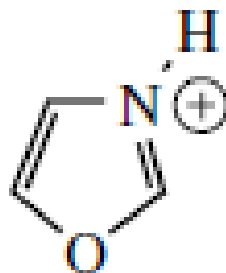


Oxazole and thiazole

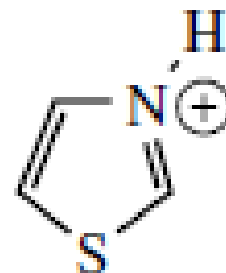
- Oxazole (pK_a 0.8) and thiazole (pK_a 2.5) are weak bases.
- The basicity of the nitrogen is reduced by the presence of the other heteroatom.
- Oxygen and sulfur provide a stronger electron-withdrawing inductive effect, compared with nitrogen, but a much weaker electron-releasing resonance effect.



pK_a 7.0



pK_a 0.8



pK_a 2.5

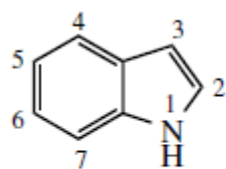
Heterocycles fused to a benzene ring

- Many interesting and important heterocyclic compounds contain fused ring systems.
- Some of the common ones are the result of fusing a heterocycle to a benzene ring, and these have long-established **trivial names**, e.g. indole, quinoline, and isoquinoline.

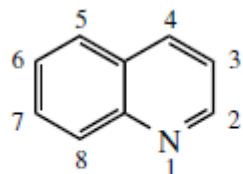
Systematic names

- can be derived by relating back to the parent heterocycle and using the prefix benzo to indicate its fusion to benzene.
- It is necessary to define which of the bonds in the heterocycle is fused to benzene,
- and this is accomplished through use of a bond descriptor,
 - a lower case italic letter in square brackets.

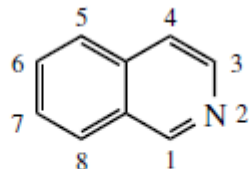
- Thus, indole is benzo[*b*]pyrrole, quinoline is benzo[*b*]pyridine, and isoquinoline, an isomer of quinoline with a different type of fusion, becomes benzo[*c*]pyridine.



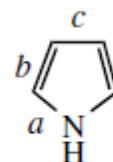
indole
benzo[*b*]pyrrole



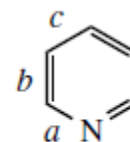
quinoline
benzo[*b*]pyridine



isoquinoline
benzo[*c*]pyridine

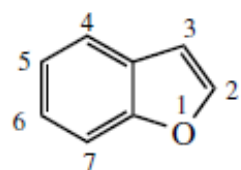


pyrrole

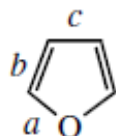


pyridine

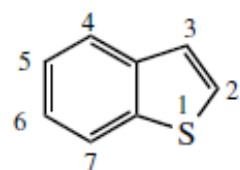
- A few other examples are shown below.
- Note that the bonds of the heterocycle are lettered starting from the heteroatom.



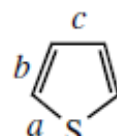
benzo[b]furan



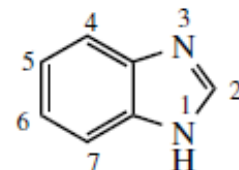
furan



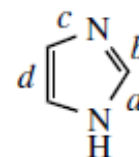
benzo[b]thiophene



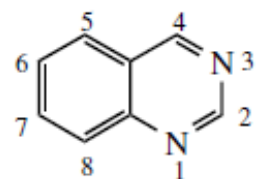
thiophene



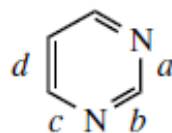
benzimidazole
benzo[d]imidazole



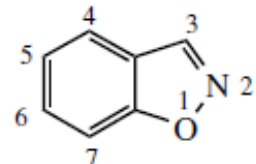
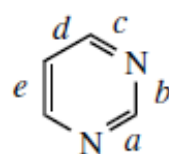
imidazole



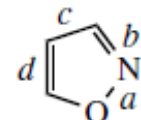
quinazoline
benzo[d]pyrimidine



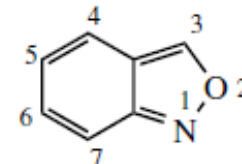
pyrimidine



benzo[d]isoxazole



isoxazole

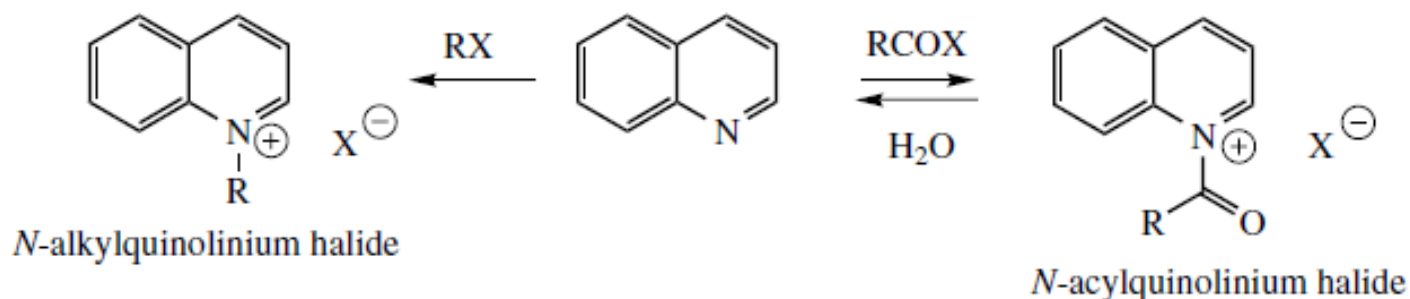


benzo[c]isoxazole

Quinoline and isoquinoline

- Quinoline and isoquinoline are **benzopyridines**.
- They behave by showing the reactivity associated with either the benzene or the pyridine rings.
- **Quinoline** is basic with a pK_a of 4.9, similar to that of pyridine (pK_a 5.2).
- As with pyridine, the nitrogen carries a lone pair in an sp^2 orbital .

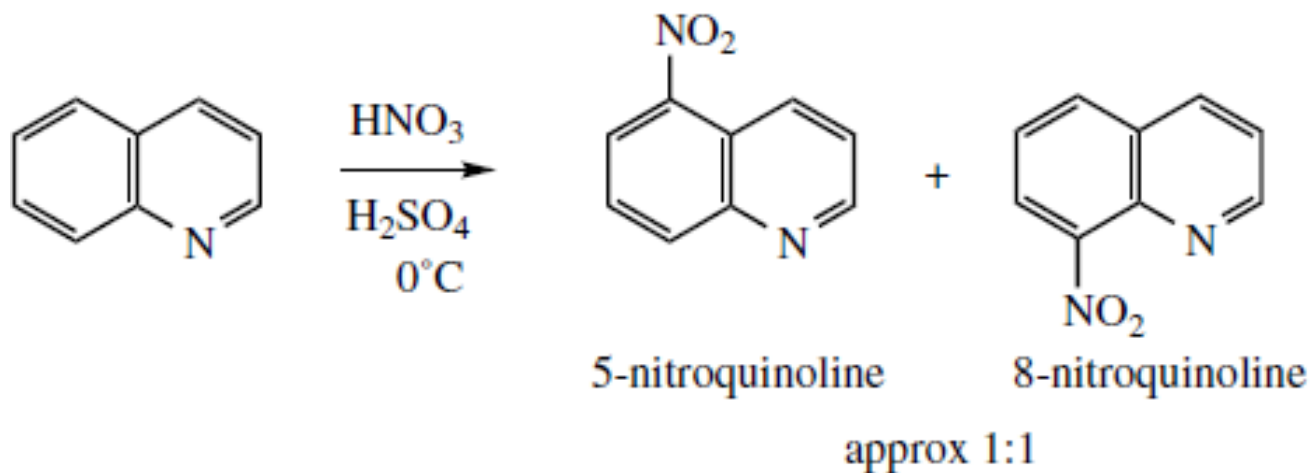
- Alkyl halides and acyl halides also react at nitrogen to give *N*-alkyl- and *N*-acylquinolinium salts.
- The *N*-alkyl salts are stable, but the *N*-acyl salts hydrolyse rapidly in the presence of water.



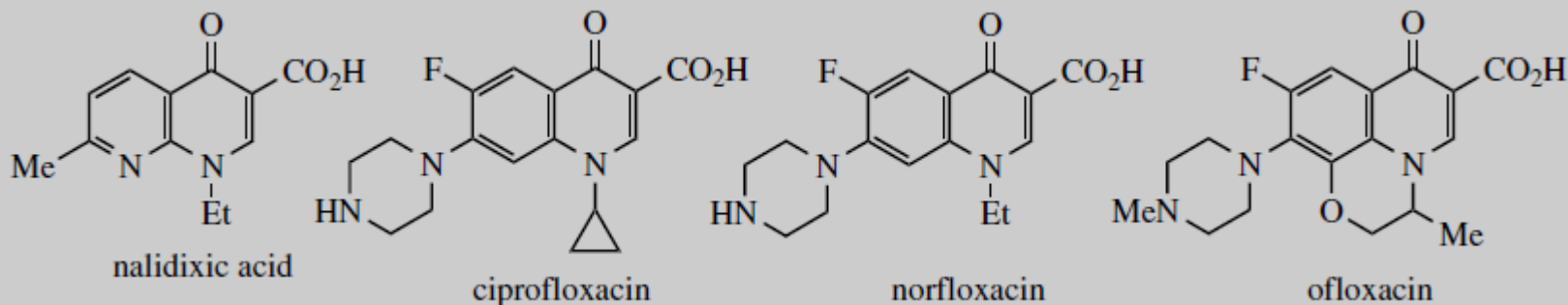
Electrophilic substitution of quinoline

- Quinoline is much more reactive towards **electrophilic substitution** than pyridine, but this is because substitution occurs on the benzene ring, not on the pyridine.
- We have already seen that pyridine carbons are unreactive towards electrophilic reagents, with strongly acidic systems protonating the nitrogen first, further inhibiting reaction .
- This is again true in quinoline, so that the protonated system is involved in the reaction, and the benzene ring undergoes substitution.

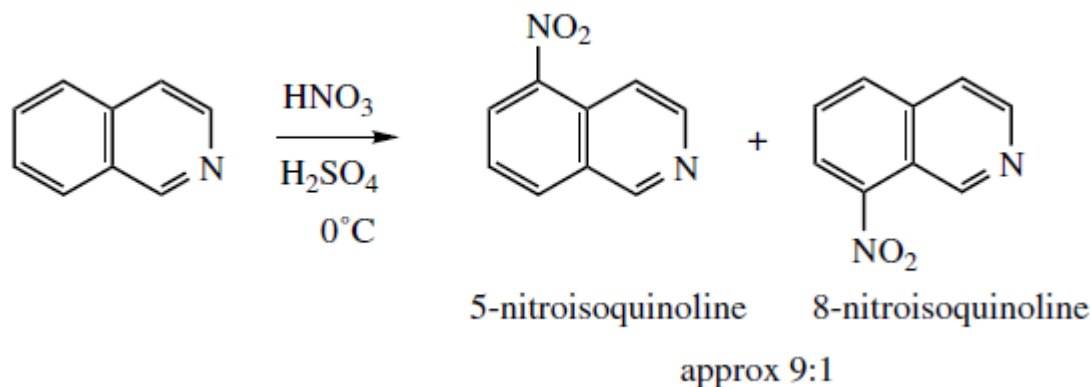
- With a nitrating mixture of HNO_3 – H_2SO_4 , the products are 5- and 8-nitroquinoline in roughly equivalent amounts.



Quinolone antibiotics



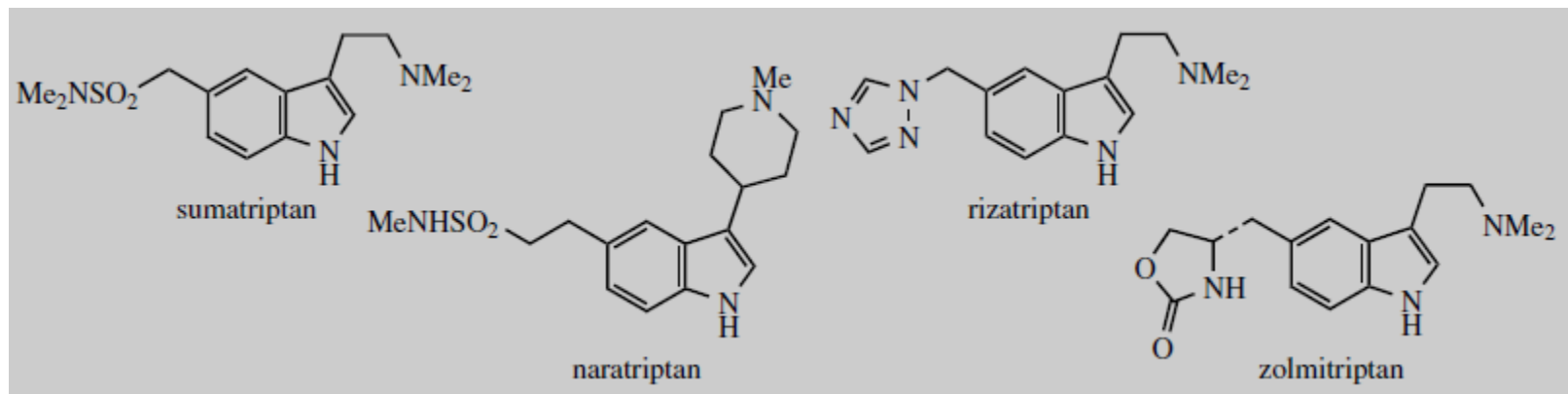
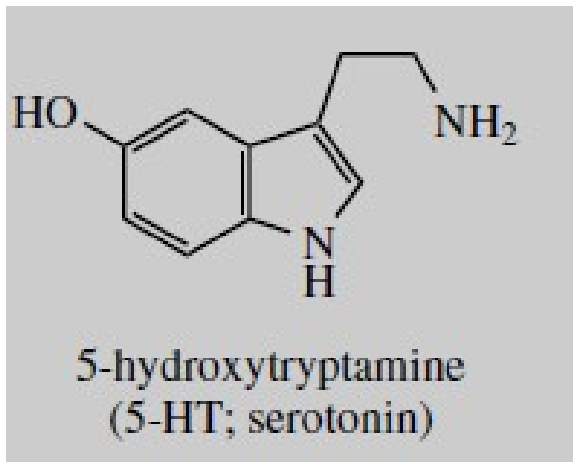
- **Isoquinoline** (pK_a 5.4)
 - has similar basicity to quinoline and pyridine, and also undergoes *N*-alkylation and *N*-acylation.
 - **Nitration** occurs smoothly to give predominantly 5-nitroisoquinoline;
 - the isoquinolinium cation reacts more readily than the quinolinium cation.



Indole

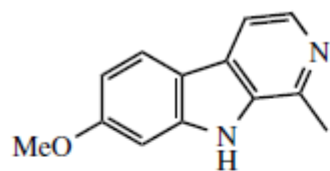
- **Indole** is the fusion of a benzene ring with a pyrrole.
- Like quinoline and isoquinoline, indole behaves as an aromatic compound.
- However, unlike quinoline and isoquinoline, where the reactivity was effectively part benzene and part pyridine, the reactivity in indole is modified by each component of the fusion.
- The closest similarity is between the chemistry of pyrroles and indoles.
- Indoles, like pyrroles, are very weak bases.
- The conjugate acid of indole has $pK_a - 3.5$; that of pyrrole has $pK_a - 3.8$.

Indoles in biochemistry



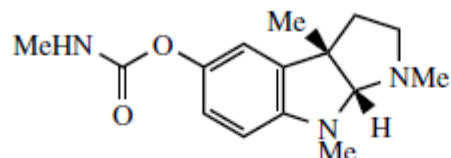
Fused heterocycles

- There is ample scope for increasing structural complexity by fusing two or more heterocycles together.
- Shown below are a few of the ring systems encountered in natural compounds, many of which have interesting, and potentially useful, biological properties.
- Note that in some cases the rings are fused so that the heteroatom can be at the ring junction and is thus common to both rings.
- This gives us even more combinations.



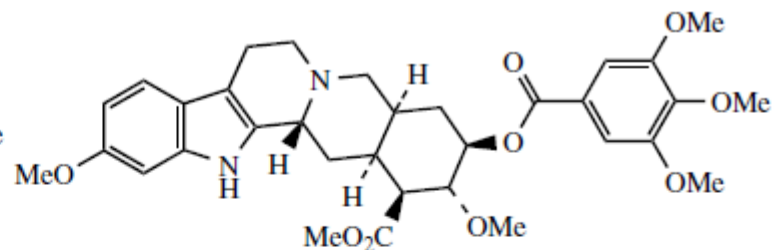
harmine

psychoactive



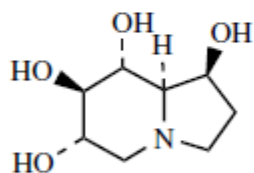
physostigmine
(eserine)

miotic; anticholinesterase



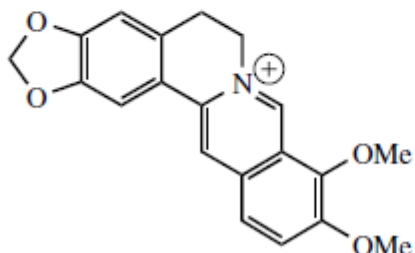
reserpine

antihypertensive

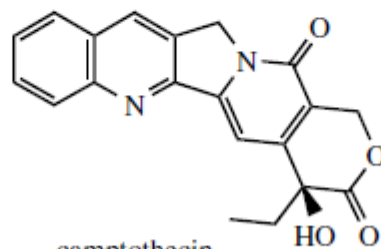


castanospermine

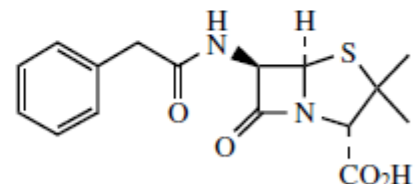
antiviral



berberine
antibacterial



camptothecin
antitumour



benzylpenicillin
(penicillin G)

antibacterial

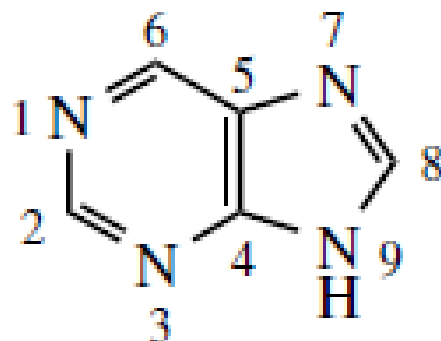
Purine and pteridine : nomenclature

- The systematic procedure for naming fused heterocycles is an extension of that we saw in a benzene ring fused to a heterocycle.
- The main difference is that we have to identify bonds in two different rings to indicate the fusion.
- We use lettering for bonds in one heterocycle and numbers for bonds in the other.
- Numbering is used for the ‘substituent’ ring and lettering for the ‘root’ ring, and all are put in square brackets between substituent and root.

Purine

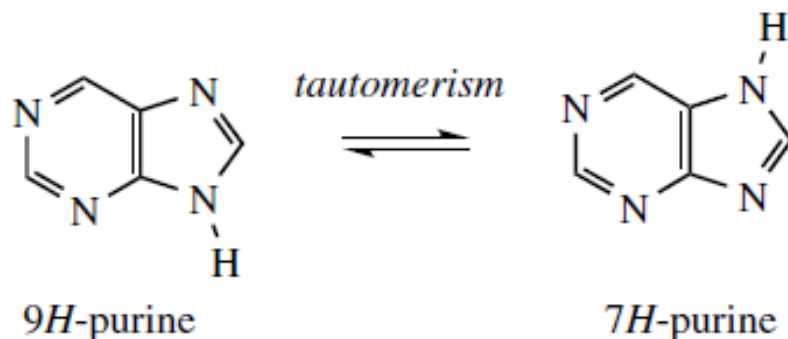
- **Purines**, along with pyrimidines , feature as bases in the nucleic acids, DNA and RNA.
- A purine is the product of fusing a five-membered imidazole ring onto a six-membered pyrimidine ring.

- The accepted numbering system unfortunately is non-systematic, and treats purine as a pyrimidine derivative, the pyrimidine ring being numbered first and separately from the other ring.



purine

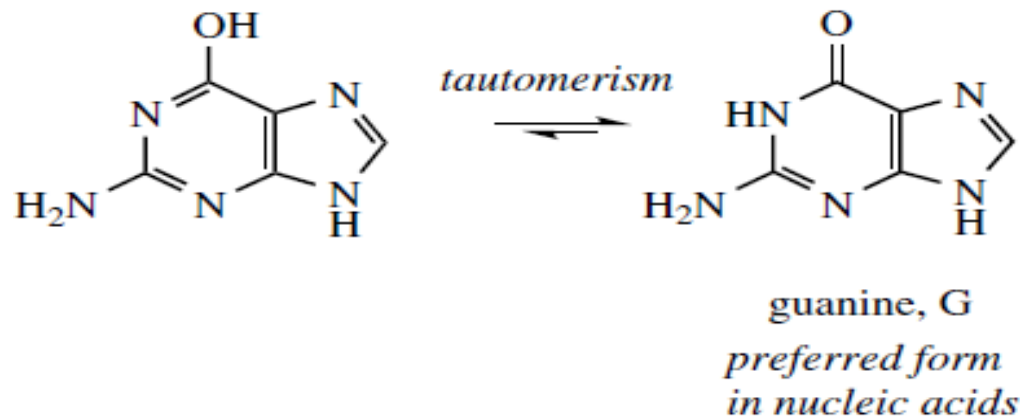
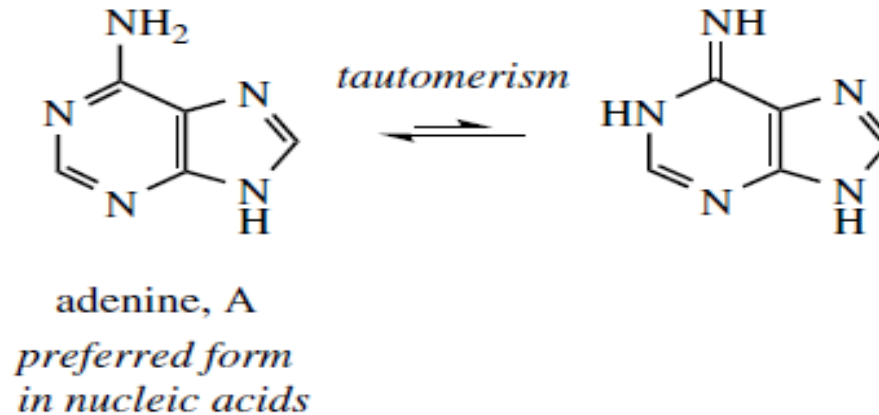
- Purine in solution exists as a roughly equimolar mixture of two tautomeric forms, *9H*-purine and *7H*purine, **tautomerism** involving the imidazole ring.



Adenine and guanine

- The purine systems in nucleic acids, adenine and guanine, are **aminopurines**.
- The amino group is on the pyrimidine ring and, as with aminopyrimidines, these compounds exist as their amino tautomers.
- Guanine also has an oxygen substituent on the pyrimidine ring, and this adopts the carbonyl form, also following the behaviour of oxypyrimidines.

- Because adenine and guanine in nucleic acids are bonded to a sugar through N-9, the additional potential for tautomerism in the imidazole ring is no longer of concern.



- Purines are quite weak bases.
- The conjugate acid of purine itself has pK_a 2.5.

Thank you